

Q1.1 Definition of structural balance [2pts]

Give a precise definition when a graph is *balanced*. Don't forget to say for what graphs balance is defined.

Un graph complet est *balanced* lorsque tous les edges (liens) ~~sont~~ forment que des triangles $+++$ ou $+- -$. Les triangles sont une propriété locale.

C'est pour les graphes complets.

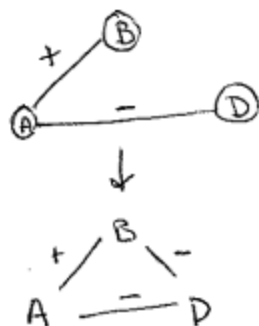
Q1.2 Structural balance theorem [3pts]

The balance theorem makes a connection between a local property and a global property for a graph. State the *balance theorem* and give a proof.

Un graph est *balanced* si il forme des triangles locaux d'uniquement $+++$ ou $+- -$; c'est *unbalanced* pour tout le reste.

Prenons un graph complet G . Soit deux sous-ensembles de ce graph X et Y . X contient que des liens d'amitié. Y ne contient que des liens d'ennemi. Les liens qui unissent X et Y sont des liens d'ennemi.

Prenons A, B un nodes du sous-ensemble X et C, D du sous-ensemble Y . $A \pm B$; $A - C$; $A - D$; $B - C$; $B - D$; $C \pm D$



Vu qu'on a ici un lien $+$ et un lien $-$; il faut que $B - D$ soient liés d'un lien d'ennemi pour que le graph garde sa propriété de *balanced*

Parce que pour A, C, D $A \pm C$ et $A \pm D$ et $C \pm D$. C sert à montrer que dans le sous-ensemble Y , C et D et les autres nodes sont liés par des liens d'amitié.

Q2 Penalty Kick game [5pts]

Given the Penalty Kick game we saw in the course with the following payoff matrix:

	Keeper		
		Dive left	Dive right
Kicker	Kick left	0.58, -0.58	0.95, -0.95
	Kick right	0.93, -0.93	0.70, -0.70

This game models penalty kicks in soccer. The payoffs are taken from an analysis of real soccer games. This game has no pure Nash equilibria, but there is a mixed-strategy Nash equilibrium.

Q2.1 Indifference principle for mixed-strategy equilibria [2pts]

Explain the principle of indifference which is used to compute mixed-strategy Nash equilibria. In your explanation, define what is a mixed-strategy equilibrium.

Le principe d'indifférence c'est quand les équilibres de Nash sont une valeur qui nous rend indifférent. Lorsque le Payoff s'annule ou est le même pour chaque stratégie dominante. Ainsi on ne peut trouver que le mixed-strategy equilibrium. *

Une mixed-strategy equilibrium est qu'il n'y a pas d'équilibre de Nash car il n'y a pas de stratégie dominante ou qu'il y a plusieurs stratégies dominantes. *

Le principe d'indifférence:
On prends cherche le meilleur Payoff
On additionnant les valeurs pour chaque
Stratégie séparément.

Q2.2 Mixed-strategy equilibrium for Penalty-Kick game [3pts]

Use the principle of indifference to compute the mixed-strategy Nash equilibrium for the Penalty-Kick game, as done in the course.

$$\text{Kick L + Dive L : } 0,58 - 0,58 \Rightarrow \text{Payoff : } 0$$

$$\text{Kick R + Dive R : } 0,70 - 0,70 \Rightarrow \text{Payoff : } 0$$

$$\text{Kick L + Dive R : } 0,95 - 0,95 \Rightarrow \text{Payoff : } 0$$

$$\text{Kick R + Dive L : } 0,93 - 0,93 \Rightarrow \text{Payoff : } 0$$

Ici alors les mixed strategy equilibria sont tous les cas possibles. Il n'y en a pas avec un payoff supérieur à 0 ni un payoff supérieur aux autres stratégies.

Q3.1 Monopolization [2pt]

Define the network structure where monopolization takes place, then give a simple example network and explain what is the equilibrium in your example. Explain why it is an equilibrium.

Monopolisation is when there is only one seller of a certain product or service in the entire network. The seller has a certain power and he can choose who to sell to or the prices for everyone.



The equilibrium in my example is either the buyer with the highest offer or if they all have the same offer it is a mixed strategy equilibrium.

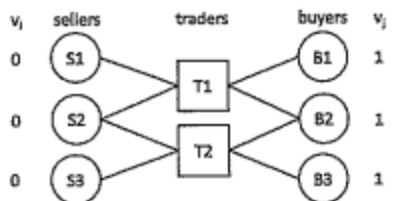
It is an equilibrium because it provides with the highest payoff to the seller and to the buyer who wants to buy.

Q3.2 Perfect competition and the indifference principle [3pt]

Define the network structure where perfect competition takes place, then give a simple example network and explain what is the equilibrium in your example. Explain why it is an equilibrium. Use the indifference principle in your explanation. Define the indifference principle for markets with intermediaries (there is a difference with Question Q2.1!).

The perfect competition is that every buyer and seller (every actor) knows everything. So this means that every actor can adapt his behaviour to be able to take the highest payoff possible for him.

Q3.3 Example trading network [3pts]



Using the monopoly and perfect competition building blocks, give the equilibria for this network and explain why they are equilibria.

The equilibrium is $S_2 - T_1 \text{ or } T_2 - B_2$ because S_2 has a better bigger number of choices than S_1 and S_3 and B_2 has also a better understanding of what the traders want. The S_2 can adapt in order to maximise payoff.

The perfect competition principle says that every S and B have the best knowledge possible so there is not a single strategy that outmatches the others in terms of knowledge but the strategy $S_2 - T_{1,2} - B_2$ provides S_2 with the choice of choosing the best trader. S_1 and S_3 can only choose 1 trader. It's the same with buyer B_2 because he can choose the best trader that suits him the most.

Q4.1 Link farm definition [2pts]

Explain what is a link farm and give the graph structure of a typical link farm.

A link farm is usually a website that lists many other websites by their links and these other websites can be link farms.

The principle hides in how the search engine of the browser works. If or when there are many links of your website in other websites it should mean that your website is important. So in repetition if many websites link many websites then there is a link farm and the websites linked receive a better PageRank score. This is no longer possible because the current version of PageRank doesn't care about the link farms. The link farms were basically an exploit of PageRank.

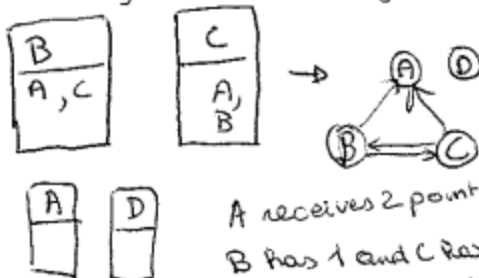
A: the website we want to promote

B, C: link farm websites

→ name of the website

→ name of the link site contains

D: is a random other website



Link farms are easier to create than real websites so even if D has links of itself in other websites, A will always have more, or create more.

D has less points than A because it is not mentioned in another website.

Q4.2 Operation of a link farm [4pts]

Using the random walk definition of Page Rank, write down the equations and use them to explain the operation of a link farm. How does the link farm achieve its effect? Could you set up a company to provide link farm services to third parties?

The random walk is the principle of randomly selecting the path to take to navigate through websites and the web.

A link farm will create infinite circles (sort of) around with other farms. So the random walker will see that there are a lot of mentions of the website and score it higher.

It is possible to set up a company to provide third party services to third parties or at least it was. All the company had to do was to create many link farm websites with many links inside every one of them. Also the random walker will have to enter the link farm from somewhere so a popular site could have a new or multiple new links added in it in order to link the farms and not to let them be disconnected from the web.

⊙ The link farms will be interconnected and make a complete graph so the random walker, once he enters in their graph, it will be hard to leave and there is a bigger chance to walk through another link farm than the actual website A from before. Therefore the infinite circles will be dependant of the number of link farms in this system.

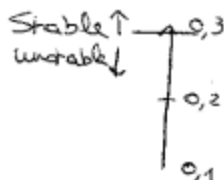
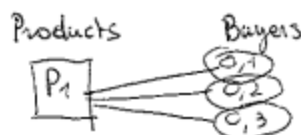
Q5.1 Market without direct-benefit effects [2pts]

In the course we defined a simple continuous model of a market without direct-benefit effects. Consumers are given names x as real numbers from 0 to 1, uniformly distributed from 0 to 1. Each consumer has a reservation price $r(x)$, arranged in decreasing order. Using a graph, explain the relationship between the price of a product and the consumers who buy the product. Explain why the production cost p^* defines a *stable equilibrium* in this market with equilibrium price x^* . Why is it stable?

$$x \in [0, 1]$$

$r(x) \rightarrow [\text{bigger} \rightarrow \text{smaller}]$

$$\text{Payoff} \equiv P = x^* - p^*$$



If $p^* < x^*$ then there is an equilibrium

If $p^* = x^*$ there must be a lesser payoff:

the payoff is zero because

$$P = x^* - p^* = p^* - p^* = 0$$

If $p^* > x^*$ then the production costs more than it

sells so it's not a good deal $P = x^* - p^*$ will be negative

Q5.2 Market with direct-benefit effects [4pts]

Extend the model of the previous question to add direct-benefit effects. Add a second function $f(z)$ that quantifies the benefit to each consumer when a fraction z of all consumers has bought the product.

- What is the formula for the *new equilibrium price* in this extended market?
- Make simple assumptions about the form of the functions $r(x)$ and $f(z)$, as we did in the course. Given these assumptions, what is the *new graph* of consumers versus price?
- What are the *equilibria* in this new graph?
- Explain for each equilibrium whether it is *stable* or *unstable* and justify your explanations.

$$f(z) = r(x) - p$$

So the benefit = the price sold - the factory cost
(the production cost)

$$r(x) = b - x$$

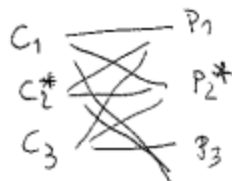
= bid - price

value that the object provides

cost of the object

$$f(z)$$

Consumers vs Price



P_2 = Price 2 is the equilibrium

because the consumers want P_2 as it provides a bigger payoff.